

Implementing RRAM in-memory computing based Spiking Neural Network

Kartik Agrawal

Bipul Biswas

D Krupakar Reddy



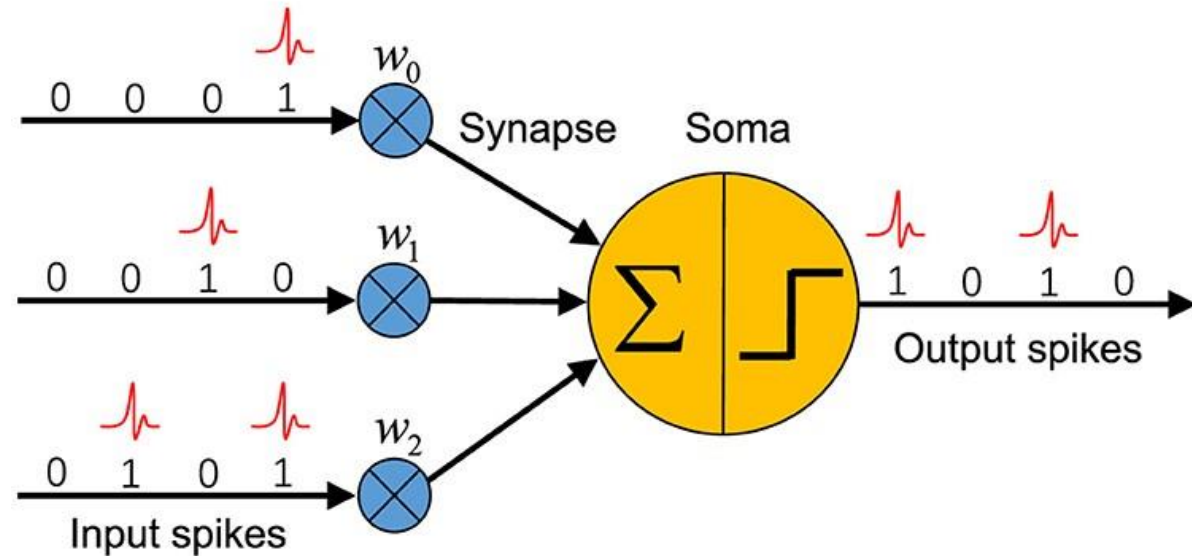
Contents

- Introduction
- Literature review
- Problem Description
- Methodology
- SW and HW implementations
- Results
- Future Prospects



Introduction - SNNs

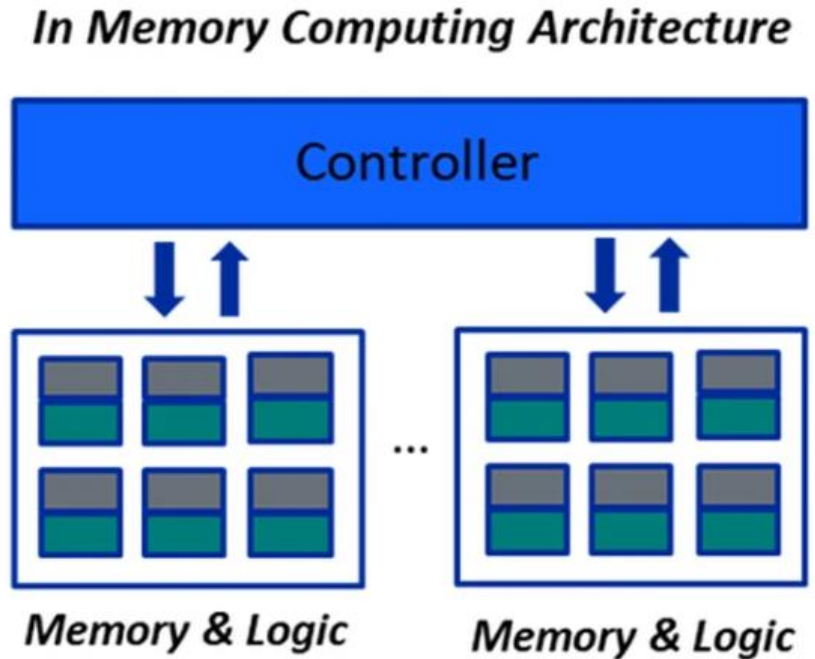
- Info encoded in timing of Spike
- Asynchronous
- Event driven
- Bio similar



Ref: Front. Neurosci., 31 July 2024 Sec. Neuromorphic Engineering Volume 18 - 2024 | <https://doi.org/10.3389/fnins.2024.1383844>

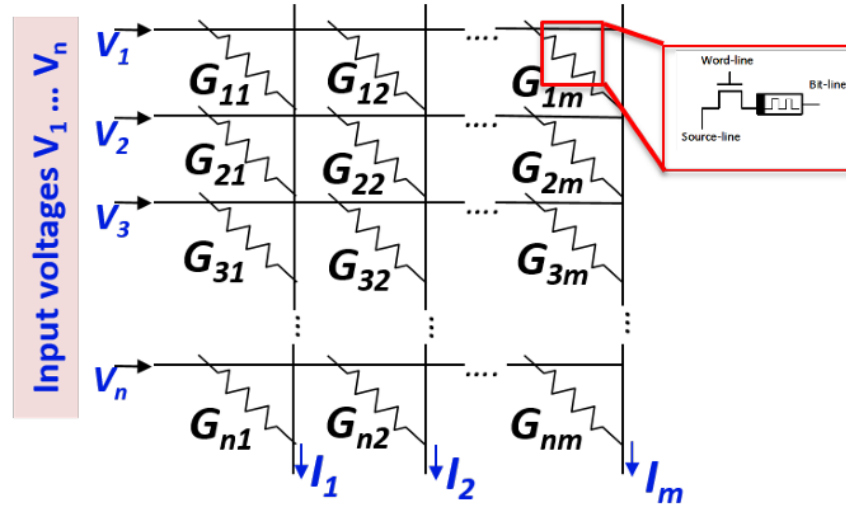
In-Memory Computing (IMC)

- Integrating memory and processing units
- Lower overheads due to data shuffling
- Parallel Processing Capabilities



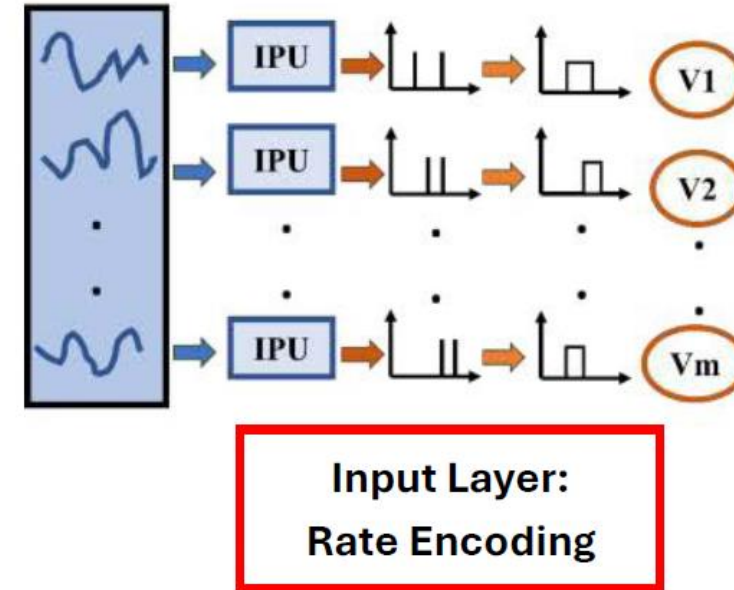
Ref: *Sci Rep* **13**, 14963 (2023). | <https://doi.org/10.1038/s41598-023-42214-x>

Literature review



1T-1R memory crossbar array

El Arrassi, Asmae, et al. "Energy-efficient SNN implementation using RRAM-based computation in-memory (CIM)." *2022 IFIP/IEEE 30th International Conference on Very Large-Scale Integration (VLSI-SoC)*. IEEE, 2022.

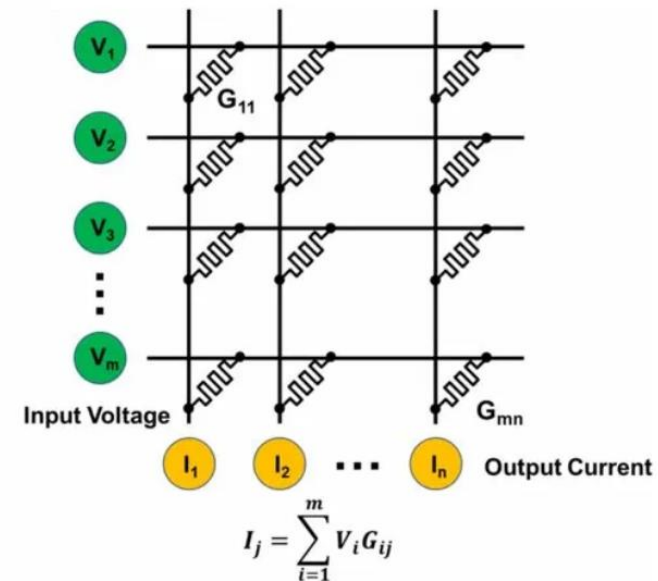
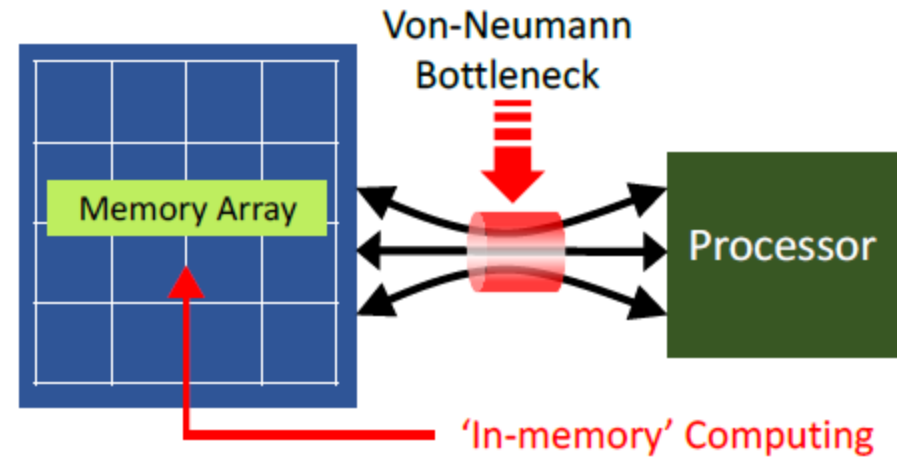


Nowshin, Fabiha, and Yang Yi. "Memristor-based deep spiking neural network with a computing-in-memory architecture." *2022 23rd International Symposium on Quality Electronic Design (ISQED)*. IEEE, 2022.

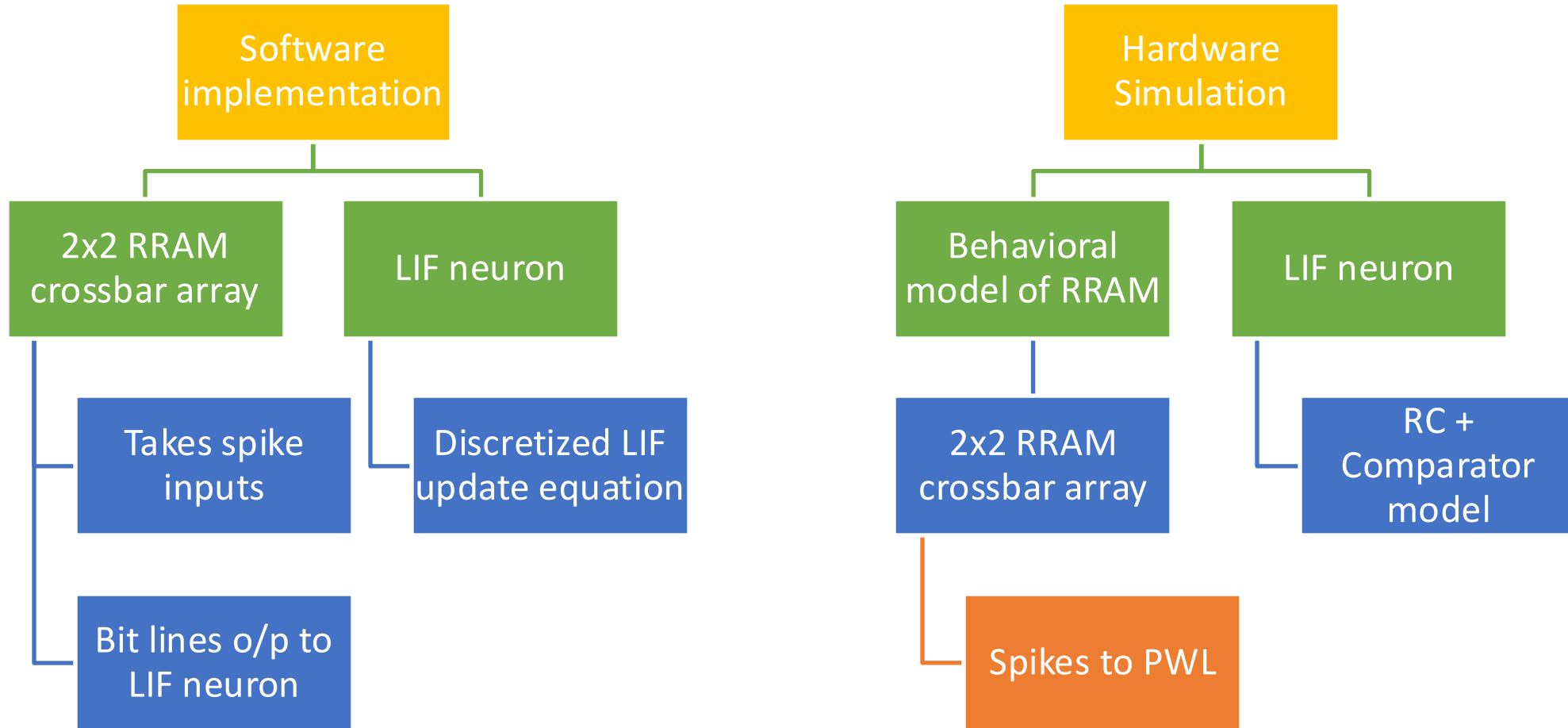


Problem Description

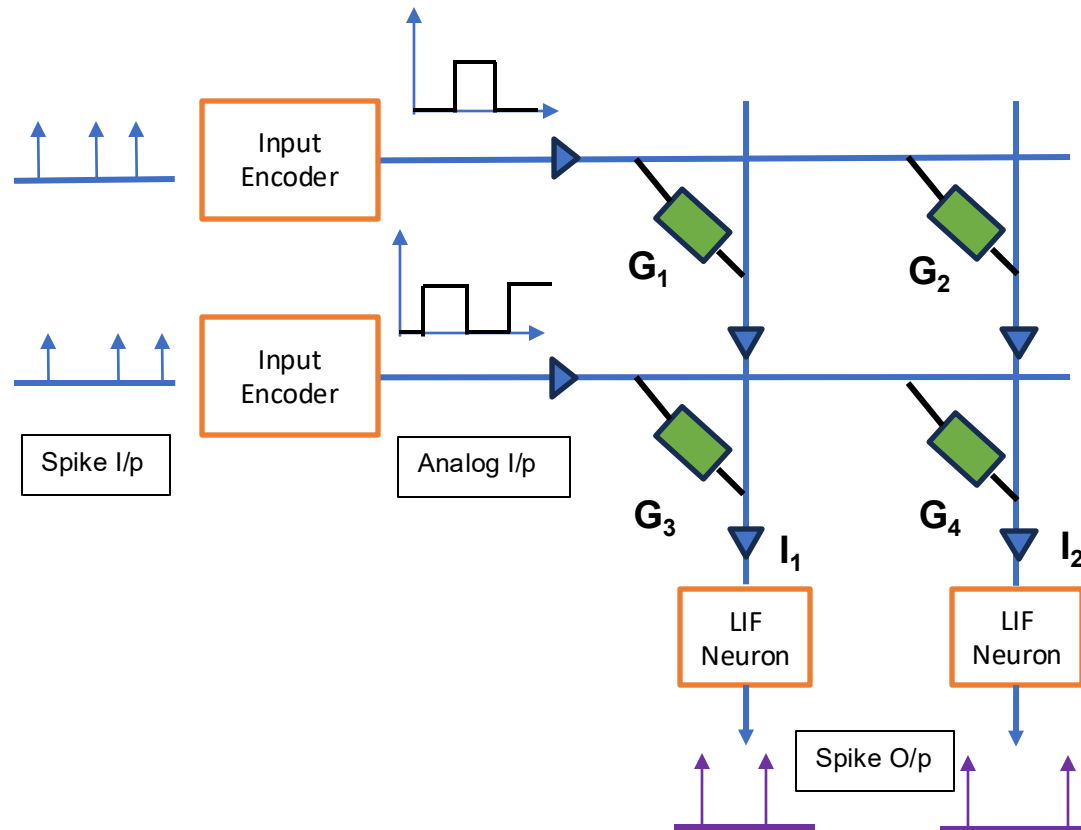
- Von-Neumann Architecture – Memory Bottle neck
- Hindering widespread adoption of ANNs in edge devices
- Energy efficient IMC capable architectures needed
- RRAM based crossbar arrays emerging as viable options



Our Methodology



Schematic



SW implementation

- 2x2 RRAM crossbar modeled in Python with spike i/p.
- It performs MAC operation and passes result as i/p to LIF neuron at end of each bit line
- LIF neuron was modeled using the discretized update equation of LIF
- Rate Encoding used for Spike train generation

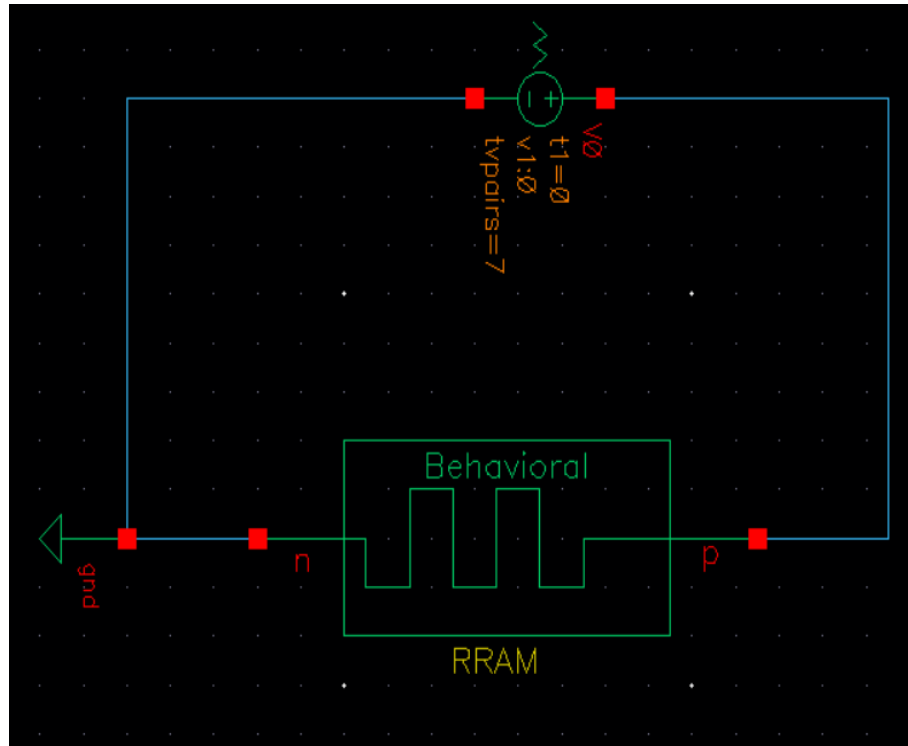
LIF Update Equation:

$$V_m[n+1] = \beta * V_m[n] + G * I[n]$$

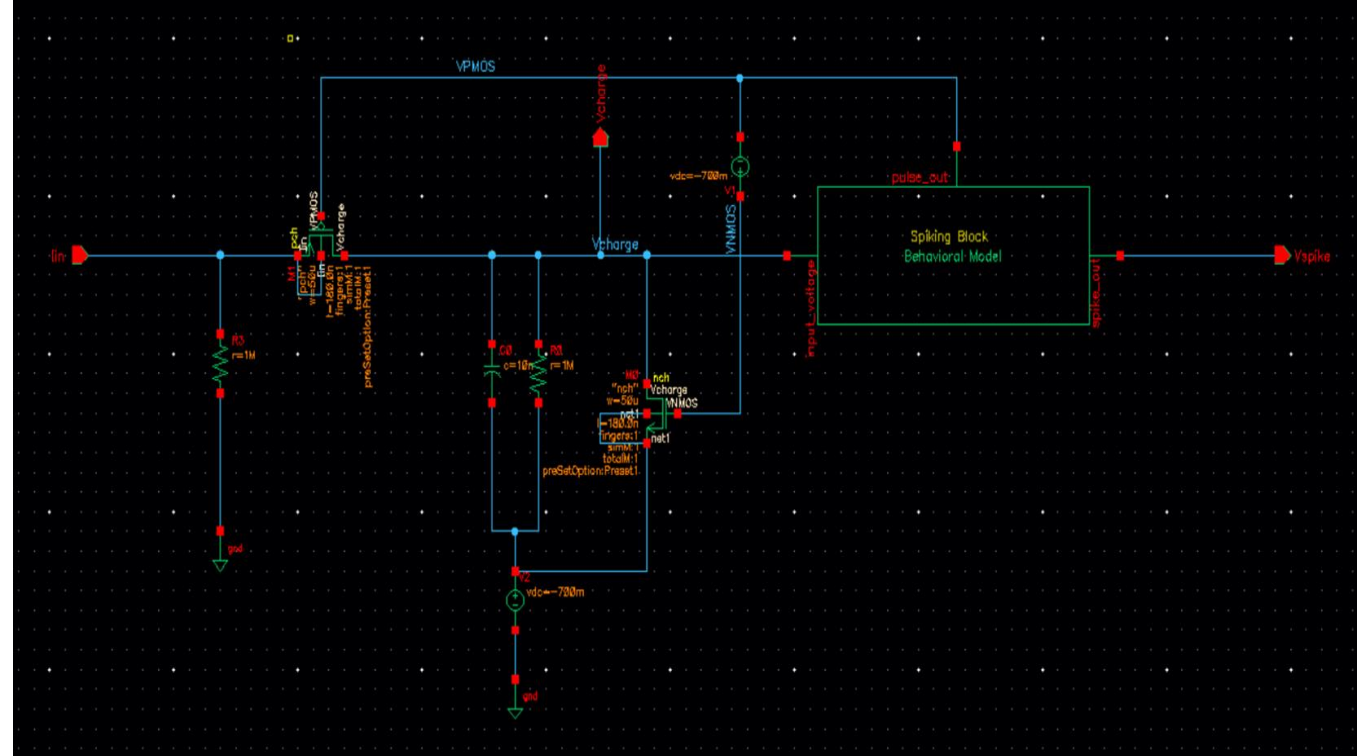
Where $\beta < 1$



Cadence Schematics



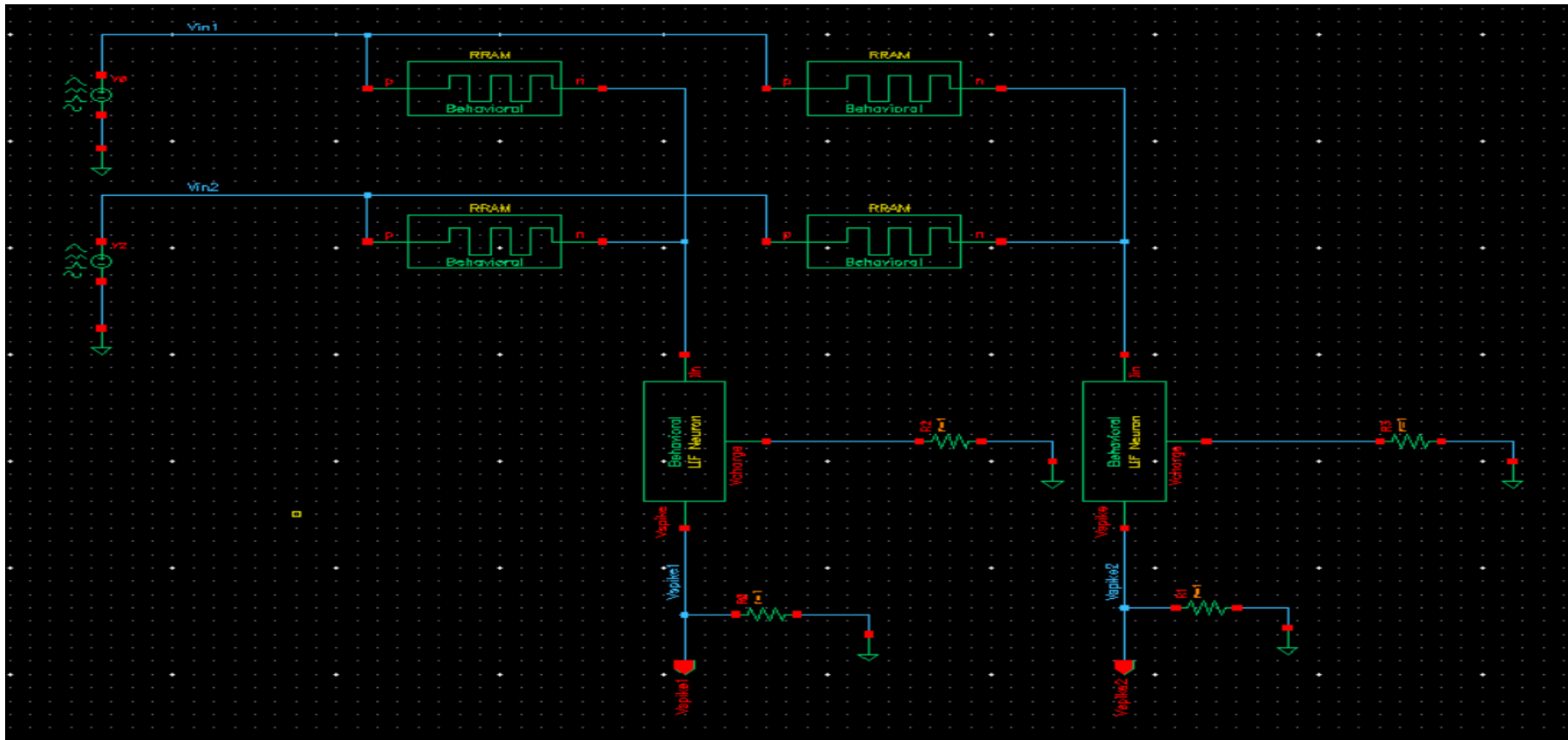
RRAM Schematic



LIF neuron Schematic



2x2 RRAM crossbar array



Setup Details

- Spike train length = 10
- Time duration of each spike = 0.5ms
- Resting potential = -0.7mV
- Threshold = -0.5mV
- $R_{\text{mem}}, C_{\text{mem}} = 500 \text{ k}\Omega, 100 \text{ pF}$
- $V_{\text{in}} = 2V$
- Conductance matrix (G) = $\begin{bmatrix} 1e-7 & 2e-7 \\ 3e-7 & 4e-7 \end{bmatrix} \text{ S}$

In order to compare the HW and SW simulation outputs, the input spike trains to both the setup were kept same

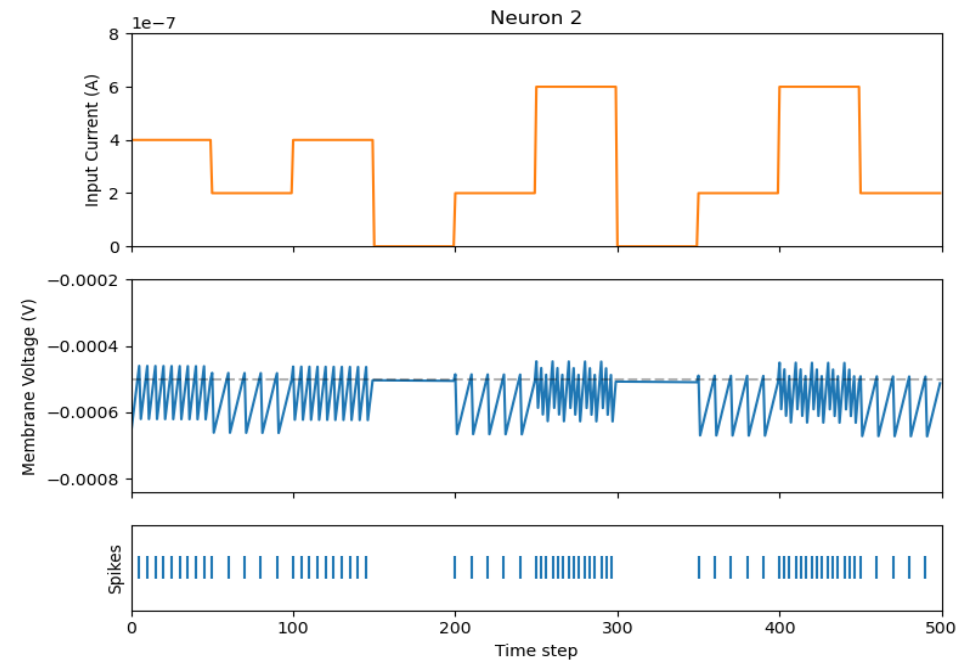
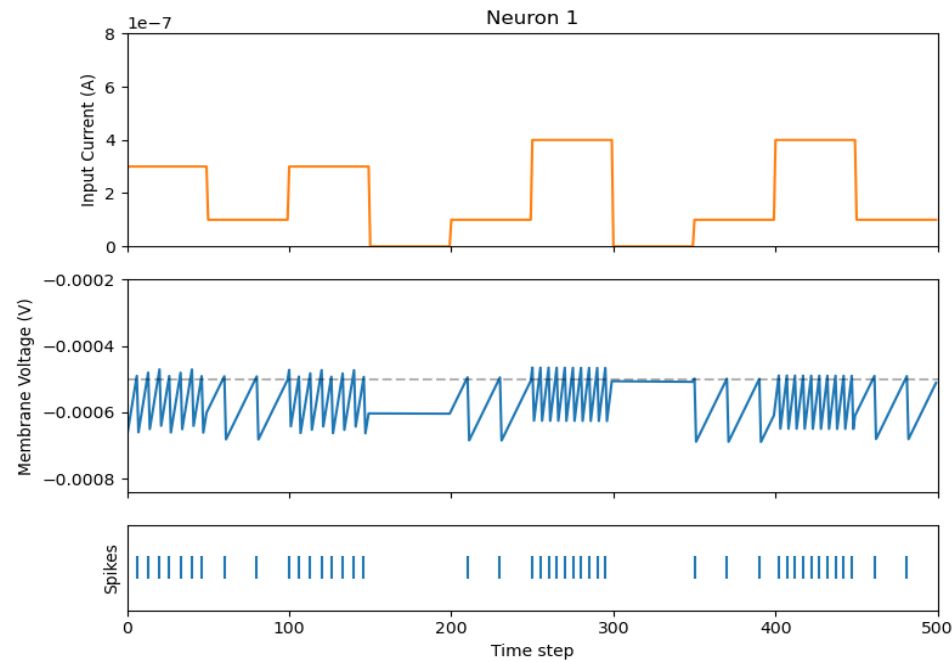
$$V_1 = [0 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1]$$

$$V_2 = [1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0]$$

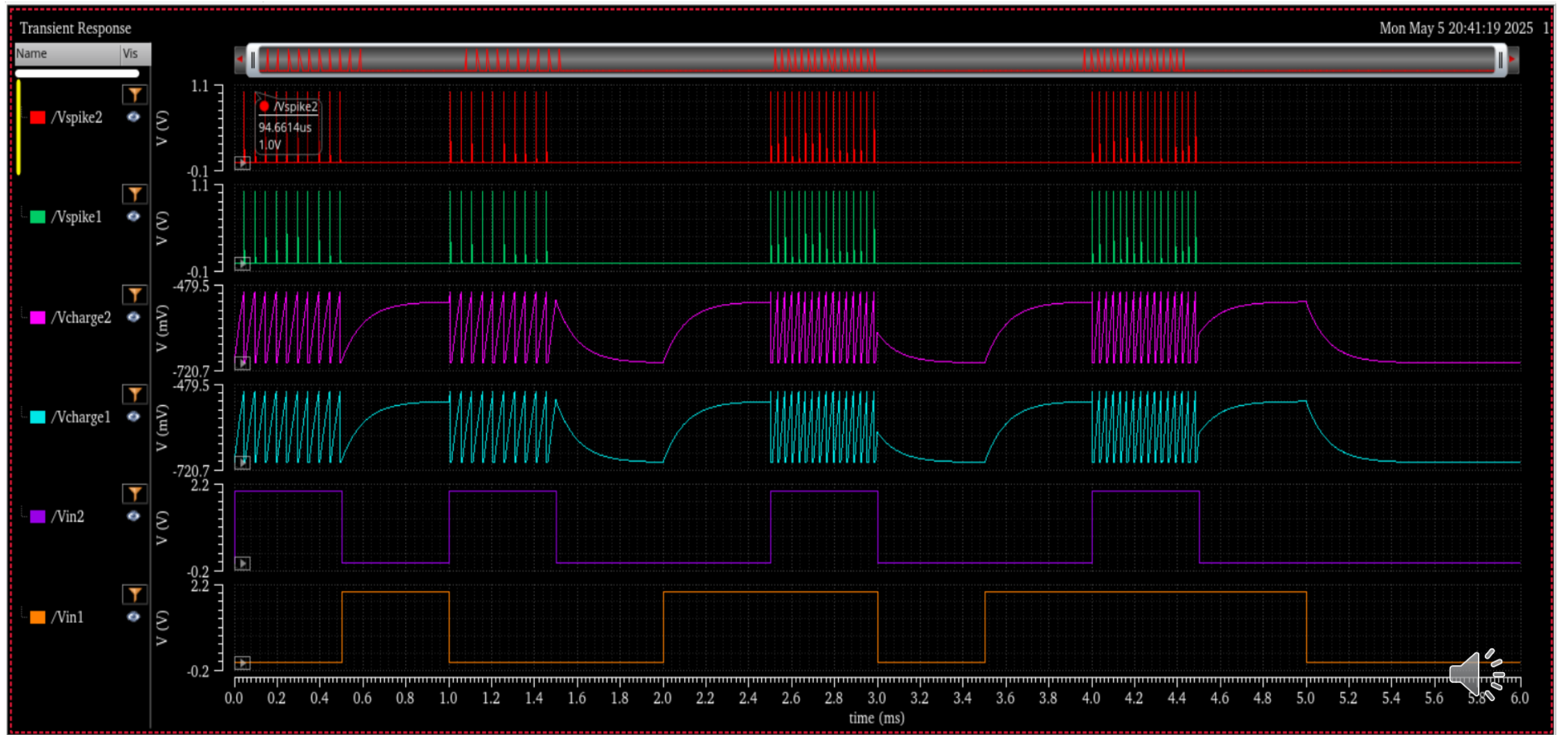


Results

Software Implementation



Hardware Simulations



Future Prospects

- Implement larger cross bar arrays for performing more complex tasks
- Implement circuit for updating conductance of RRAMs through STDP (on-site learning)
- Circuit for inter spike interval encoding
- Implement online learning capabilities

